

SYDNEY TECHNICAL HIGH SCHOOL
(Est 1911)

MATHEMATICS EXTENSION 2

HSC ASSESSMENT TASK 2

JUNE 2003

Time allowed : 70 minutes

Instructions :

- Show all necessary working in every question.
- Start each question on a new page.
- Attempt all questions.
- All questions are of equal value.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- This test forms part of your HSC assessment.
- These questions are to be handed in with your answers.

Name : _____

Question 1	Question 2	Question 3	Total

Question 1

i) Find the following indefinite integrals

a. $\int \frac{6 \, dx}{x^2 + 5x + 4}$ by using partial fractions 3

b. $\int \frac{2x \, dx}{x^2 + 4x + 7}$ 4

ii) Fully factorise $x^3 - 4x^2 + 7x - 6$ over the complex field 2

iii) The point $P(a \sec \theta, b \tan \theta)$ lies on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

a) Derive the equation of the tangent to the hyperbola at $P(a \sec \theta, b \tan \theta)$. 3

b) Show that the coordinates of A ,

the point where the tangent cuts the y axis are $(0, \frac{-b}{\tan \theta})$ 1

c) Given that the normal to the hyperbola at $P(a \sec \theta, b \tan \theta)$ is given by

$ax \tan \theta + by \sec \theta = (a^2 + b^2) \sec \theta \tan \theta$ find the coordinates of the point B , 1

the point where the normal cuts the y axis.

d) Show that a focus of the hyperbola lies on a circle with diameter AB . 3

Question 2 (Start a new page)

i) Solve the equation $x^4 - 6x^3 + 12x^2 - 10x + 3 = 0$ 3

given that it has a triple root.

ii) $T(ct, \frac{c}{t})$ is a point on the hyperbola $xy = c^2$.

a) Show that the equation of the tangent to the hyperbola $xy = c^2$ 3

at the point $T(ct, \frac{c}{t})$ is given by $x + t^2y = 2ct$.

b) This tangent meets the x and y axes at A and B respectively.

The point C is the fourth vertex of the rectangle $AOBC$ where O is the origin. 4

Find the locus of C .

iii) Let ω be the complex root of $z^3 - 1 = 0$ with the least positive argument.

a) Explain why $1 + \omega + \omega^2 = 0$. 1

b) Find in its simplest form, the polynomial equation with roots

$a + b$, $a\omega + b\omega^2$ and $a\omega^2 + b\omega$ where a and b are real 3

iv) The polynomial $P(x)$ is given by $P(x) = 2x^3 - 9x^2 + 12x - k$, 3

where k is real.

Using calculus or otherwise, find all the possible values of k if $P(x) = 0$

has exactly one real root and two complex roots.

Question 3 (Start a new page)

i) Find the following indefinite integrals

a) $\int \frac{e^x dx}{\sqrt{2-e^{2x}}}$ 2

b) $\int \sin^5 x \, dx$ 3

ii) If α , β and δ are the roots of the equation $x^3 + qx + r = 0$

find the polynomial equation with roots ;

a) $\alpha - 2$, $\beta - 2$ and $\delta - 2$. 2

b) $\alpha\beta$, $\alpha\delta$ and $\beta\delta$. 3

iii) When the polynomial $P(x)$ is divided by $x - i$ the remainder equals $-3 - i$. 3

Find the remainder when $P(x)$ is divided by $x^2 + 1$.

iv) The polynomial $P(x)$ is defined by the rule $P(x) = x^4 + Ax^2 + B$

where A and B are positive real integers.

a) If two of the zeroes of $P(x)$ are bi and di where b and d are real, 1

state the other two zeroes in terms of b and d .

b) Show that $b^4 + d^4 = A^2 - 2B$ 3

Question 1

$$i) a. \frac{6}{x^2+5x+4} = \frac{A}{x+4} + \frac{B}{x+1}$$

$$\therefore 6 = A(x+1) + B(x+4)$$

$$\text{let } x = -1 \quad 6 = 3B$$

$$\therefore B = 2$$

$$\text{let } x = -4 \quad 6 = -3A$$

$$A = -2$$

$$\begin{aligned} \int \frac{6 dx}{x^2+5x+4} &= \int \frac{2}{x+1} - \frac{2}{x+4} dx \\ &= 2 \ln(x+1) - 2 \ln(x+4) \\ &= 2 \ln\left(\frac{x+1}{x+4}\right) \end{aligned}$$

$$\begin{aligned} b. \int \frac{2x dx}{x^2+4x+7} &= \int \frac{2x+4}{x^2+4x+7} dx - \int \frac{4 dx}{x^2+4x+7} \\ &= \ln(x^2+4x+7) - \int \frac{4 dx}{(x+2)^2+3} \\ &= \ln(x^2+4x+7) - \frac{4}{\sqrt{3}} \tan^{-1} \frac{x+2}{\sqrt{3}} \end{aligned}$$

$$ii) P(2) = 0 \quad \therefore x-2 \text{ is a factor}$$

$$\begin{aligned} \therefore (x-2)(x^2-2x+3) &= (x-2)(x^2-2x+1+2) \\ &= (x-2)((x-1)^2+2) \\ &= (x-2)(x-1+\sqrt{2}i)(x-1-\sqrt{2}i) \end{aligned}$$

$$iii) a. \frac{2x}{a^2} - \frac{2y}{b^2} \cdot \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{b^2 x}{a^2 y}$$

$$\text{sub } P(a \sec \theta, b \tan \theta)$$

$$m_T = \frac{b^2 a \sec \theta}{a^2 b \tan \theta}$$

$$= \frac{b \sec \theta}{a \tan \theta}$$

$\therefore$ equation

$$y - b \tan \theta = \frac{b \sec \theta}{a \tan \theta} (x - a \sec \theta)$$

$$ay \tan \theta - ab \tan^2 \theta = bx \sec \theta - ab \sec^2 \theta$$

$$bx \sec \theta - ay \tan \theta = ab(-\tan^2 \theta + \sec^2 \theta)$$

$$bx \sec \theta - ay \tan \theta = ab$$

$$\frac{x \sec \theta}{a} - \frac{y \tan \theta}{b} = 1$$

$$b. \text{ when } x=0$$

$$-\frac{y \tan \theta}{b} = 1$$

$$y = -\frac{b}{\tan \theta}$$

$$\therefore A\left(0, -\frac{b}{\tan \theta}\right)$$

$$c. \text{ when } x=0$$

$$by \sec \theta = (a^2 + b^2) \sec \theta \tan \theta$$

$$y = \frac{(a^2 + b^2) \tan \theta}{b}$$

$$\therefore B\left(0, \frac{(a^2 + b^2) \tan \theta}{b}\right)$$

d. focus $S(ae, 0)$

$$m_{AS} = \frac{\frac{b}{\tan \theta}}{ae}$$

$$= \frac{b}{ae \tan \theta}$$

$$m_{BS} = \frac{\frac{(a^2+b^2) \tan \theta}{b}}{-ae}$$

$$= \frac{(a^2+b^2) \tan \theta}{-abe}$$

$$\therefore m_{AS} + m_{BS} = \frac{b}{ae \tan \theta} \times \frac{(a^2+b^2) \tan \theta}{-abe}$$

$$= -\frac{(a^2+b^2)}{(ae)^2}$$

$$\text{but } b^2 = a^2(e^2 - 1)$$

$$a^2 e^2 = a^2 + b^2$$

$$\therefore m_{AS} + m_{BS} = -1$$

$\therefore$ AB subtends a right angle at S

$\therefore$ S lies on circle with diameter AB.

Question 2

$$i) \quad x^4 - 6x^3 + 12x^2 - 10x + 3 = 0$$

$$4x^3 - 18x^2 + 24x - 10 = 0$$

$$12x^2 - 36x + 24 = 0$$

$$12(x-2)(x-1) = 0$$

$$x = 1, 2$$

by inspection triple root is $x=1$

($x=2$ does not satisfy equation)

$\therefore$ sum of roots

$$1+1+1+\alpha = 6$$

$$\therefore \alpha = 3$$

$\therefore$ Solutions: 1, 1, 1, 3

$$ii) a. \quad y = c^2 x^{-1}$$

$$\frac{dy}{dx} = -c^2 x^{-2}$$

$$\text{when } x = ct$$

$$m_1 = \frac{-c^2}{(ct)^2}$$

$$= -\frac{1}{t^2}$$

$\therefore$ equation

$$y - \frac{c}{t} = -\frac{1}{t^2}(x - ct)$$

$$t^2 y - ct = -x + ct$$

$$x + t^2 y = 2ct$$

$$b. \quad \text{when } y=0 \quad x=2ct$$

$$\therefore A(2ct, 0)$$

$$\text{when } x=0 \quad t^2 y = 2ct$$

$$y = \frac{2c}{t}$$

$$\therefore B(0, \frac{2c}{t})$$

$$\therefore C(2ct, \frac{2c}{t})$$

lows

$$\therefore x=2ct, \quad y = \frac{2c}{t}$$

||

$$t = \frac{x}{2c}$$

$$\therefore y = \frac{2c}{\frac{x}{2c}}$$

$$\underline{xy = 4c^2}$$

iii) a) w is a root of $z^3 - 1 = 0$
 $\therefore$ other roots are $1, w^2$
 $\therefore 1 + w + w^2 = 0$ sum of roots

b) $a+b, aw+bw^2, aw^2+bw$

$$\begin{aligned} \therefore S_1 &= a+b+aw+bw^2+aw^2+bw \\ &= a+b+a(w+w^2)+b(w+w^2) \\ &= a+b-a-b \\ &= 0 \end{aligned}$$

$$\begin{aligned} S_2 &= (a+b)(aw+bw^2) + (a+b)(aw^2+bw) \\ &\quad + (aw+bw^2)(aw^2+bw) \\ &= a^2(w+w^2+w^3) + b^2(w^2+w+w^3) \\ &\quad + ab(w^2+w+w+w^2+w^2+w^4) \\ &= 3ab(w+w^2) \\ &= -3ab \end{aligned}$$

$$\begin{aligned} S_3 &= (a+b)(aw+bw^2)(aw^2+bw) \\ &= (a+b)(a^2w^3+abw^2+abw^4+b^2w^3) \\ &= (a+b)(a^2-ab+b^2) \\ &= a^3+b^3 \end{aligned}$$

$\therefore$ equation is

$$x^3 - 3abx - (a^3+b^3) = 0$$

iv) $p'(x) = 6x^2 - 18x + 12$
 $= 6(x-2)(x-1)$

$\therefore$ Turning pts at $x=1, 2$

both turning pts need to be on same side of x axis

$p(1) = 5-k$ (max)

$p(2) = 4-k$ (min)

$$\begin{aligned} \therefore 4-k > 0 \quad \text{or} \quad 5-k < 0 \\ \underline{k < 4} \quad \text{or} \quad \underline{k > 5} \end{aligned}$$

Question 3

i) a. let $u = e^x$
 $du = e^x dx$

$$\begin{aligned} \therefore \int \frac{du}{\sqrt{2-u^2}} \\ &= \sin^{-1} \frac{u}{\sqrt{2}} \\ &= \sin^{-1} \left(\frac{e^x}{\sqrt{2}} \right) \end{aligned}$$

b. $\int \sin^5 x dx$

$$\begin{aligned} &= \int \sin^4 x \sin x dx \\ &= \int (1 - \cos^2 x)^2 \sin x dx \\ &\quad \text{let } u = \cos x \\ &\quad du = -\sin x dx \end{aligned}$$

$$\begin{aligned} &= - \int (1-u^2)^2 du \\ &= - \int 1 - 2u^2 + u^4 du \\ &= - \left[u - \frac{2}{3}u^3 + \frac{1}{5}u^5 \right] \\ &= -\cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x \end{aligned}$$

ii) a. $p(x+2) = 0$

$$\begin{aligned} \therefore (x+2)^3 + q(x+2) + r &= 0 \\ x^3 + 6x^2 + 12x + 8 + qx + 2q + r &= 0 \\ x^3 + 6x^2 + (12+q)x + 2q+r+8 &= 0 \end{aligned}$$

b. $\alpha\beta, \alpha\delta, \beta\delta$
 $\Rightarrow \frac{\alpha\beta\delta}{\delta}, \frac{\alpha\beta\delta}{\beta}, \frac{\alpha\beta\delta}{\alpha}$
 $\alpha\beta\delta = -r$

$$\Rightarrow \frac{-r}{\alpha}, \frac{-r}{\beta}, \frac{-r}{\delta}$$

$$\therefore P\left(\frac{-r}{x}\right) = 0$$

$$\left(\frac{-r}{x}\right)^3 + q\left(\frac{-r}{x}\right) + r = 0$$

$$\frac{-r^3}{x^3} - \frac{qr}{x} + r = 0$$

$$-r^3 - qr x^2 + r x^3 = 0$$

$$r x^3 - q r x^2 - r^3 = 0 \quad \underline{\div r}$$

$$x^3 - q x^2 - r^2 = 0$$

$$\text{iii) } P(x) = (x^2 + 1)Q(x) + ax + b$$

$$\text{but } P(i) = -3 - i$$

$$\therefore -3 - i = 0 + ai + b$$

$$a = -1 \quad b = -3$$

$$\therefore \text{remainder} = -x - 3$$

$$\text{v) } a. -bi, -di$$

$$b. \quad b^4 + d^4 = (b^2 + d^2)^2 - 2b^2d^2$$

taking roots 2 at a time

$$b^2 - bd + bd + bd - bd + d^2 = A$$

$$\therefore b^2 + d^2 = A$$

taking roots 4 at a time

$$(-bi)(-di)(bi)(di) = B$$

$$\therefore b^2d^2 = B$$

$$\therefore b^4 + d^4 = (b^2 + d^2)^2 - 2b^2d^2$$

$$= A^2 - 2B$$